A- HOME LAB DEPLOYMENT IN VMWARE WORKSTATION17PRO.

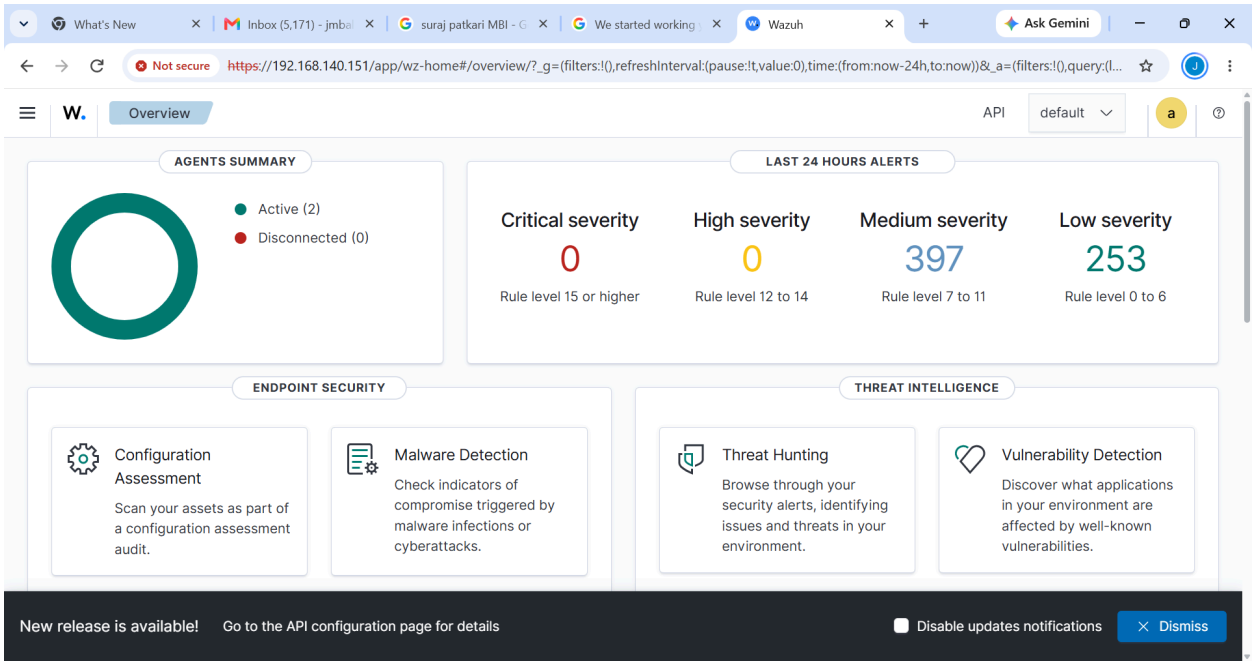
- 192.168.140.X/24 Network

*Kali Linux (attacker's endpoint) 192.168.140.148
*WIN10Pro (victim's endpoint) 192.168.140.147
*Ubuntu server 192.168.140.150
*WAZUH SIEM 192.168.140.151

There has been a use of AI in assisting in some of the steps in completing these labs on troubleshooting and to clear some bottlenecks.

1- WAZUH SIEM DEPLOYMENT ON UBUNTU 22.04Lts and WIN10PRO.

This lab project is about deployment of an enterprise-level WAZUH SIEM/EDR to aggregate local telemetries from virtual endpoints in a VMware installation. Then install an indexer in an UBUNTU server 22.04, to monitor events in the VMware hypervisor from Win10Pro VM and an agent locally installed on the UBUNTU server to send events to the Wazuh SIEM dashboard.

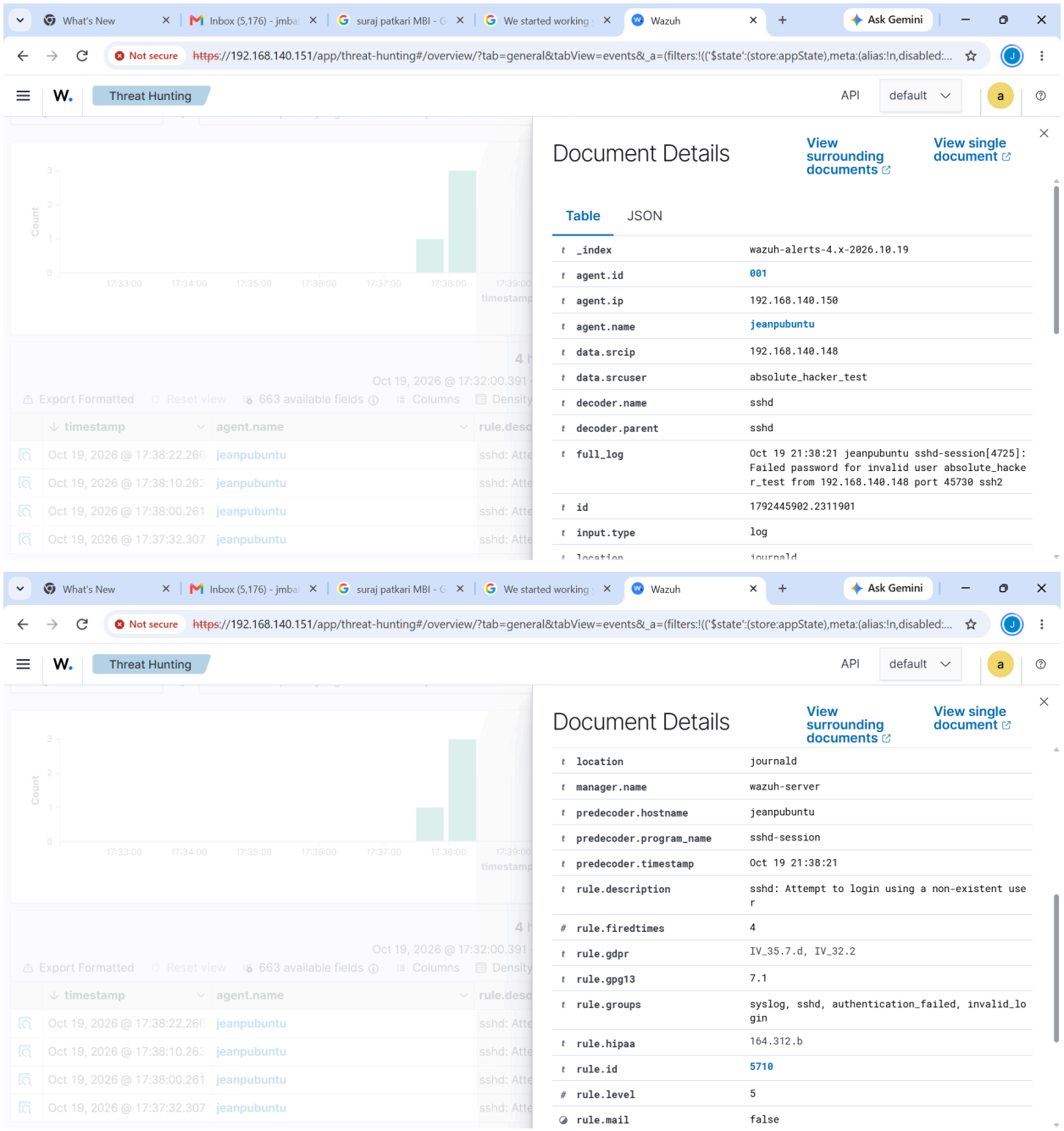


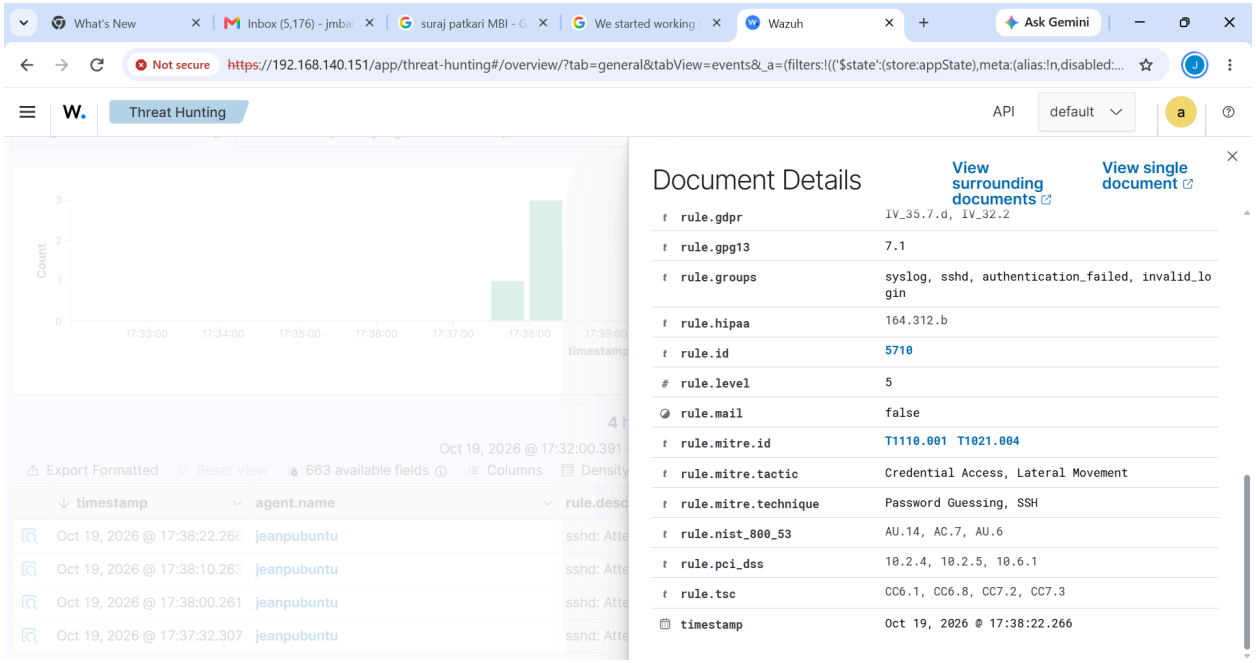
a - Wazuh dashboard screenshot overview after deployment and browser launch.

This screenshot shows two endpoints "active" with a summary of alerts by severity: Medium(397) and Low(253) identified by the agents. Moreover, you can see a "configuration assessment", a "Malware detection", a "Threat hunting" and a "Vulnerability management" sections.

A simulation lab example of an intrusion attack followed by Threat hunting in the SIEM details "table" section through the SSH protocol is shown below from an alert matching a SIEM rule in

Ubuntu server(192.168.140.150). It gives the description of the alert, the source of the action, the destination and the proof of activity in a detailed display.



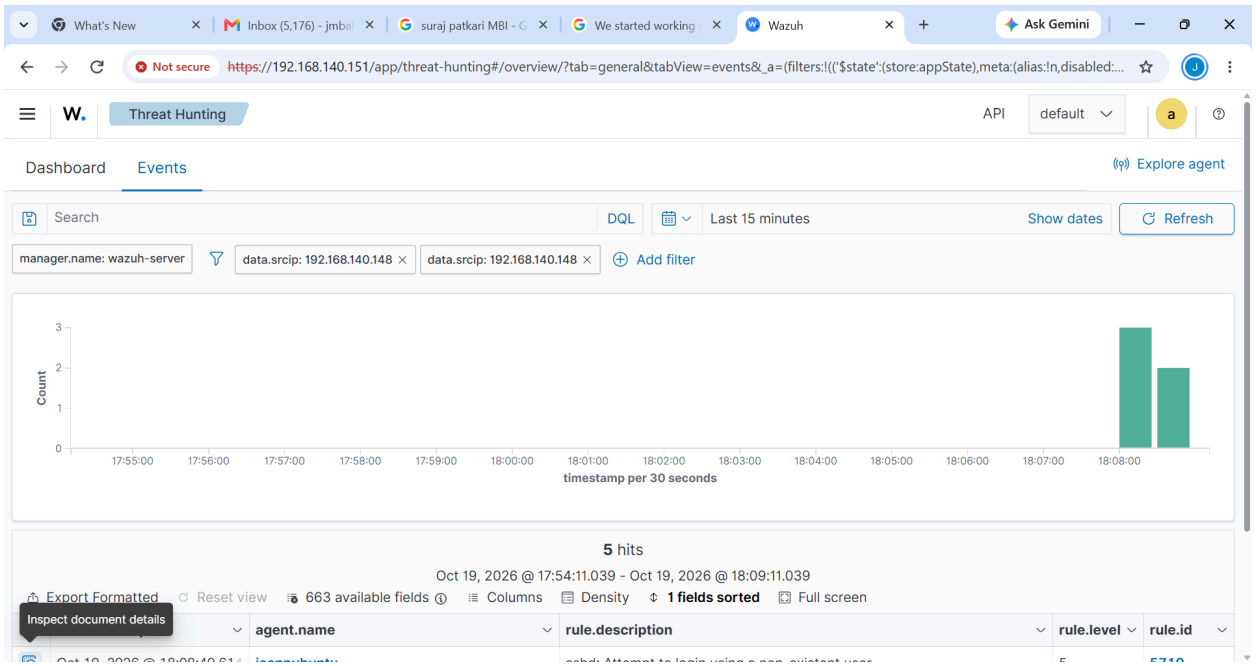


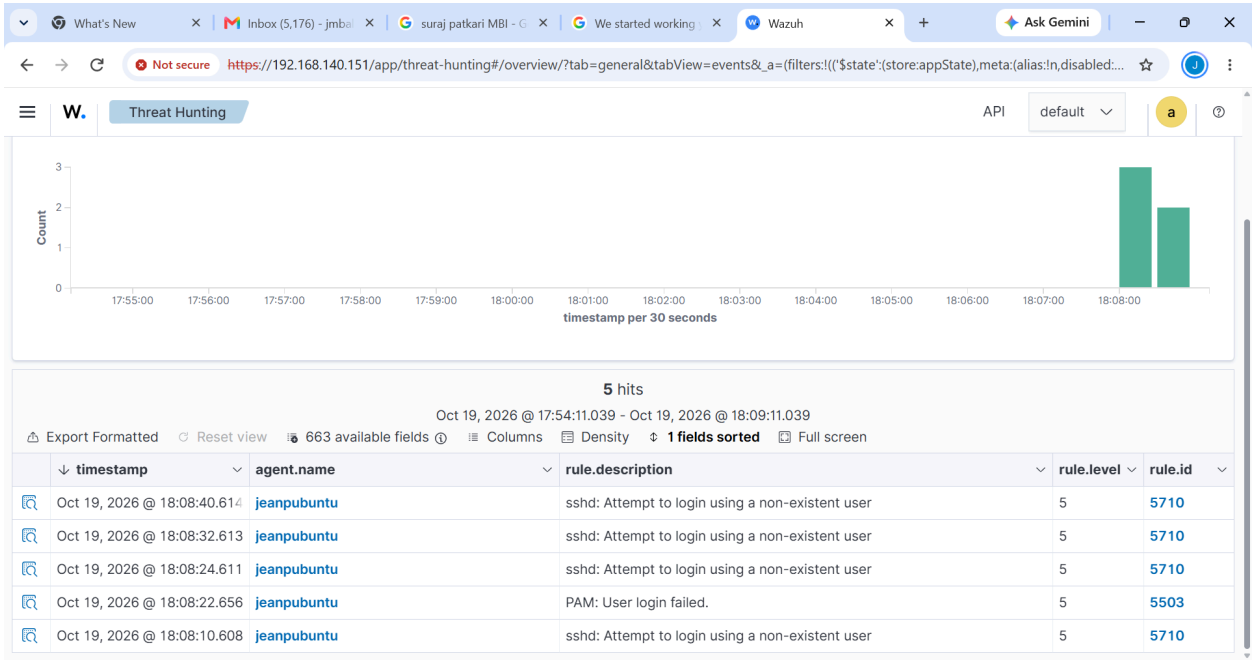
b- Screenshots of a detailed table of the alert as indexed in the SIEM with description.

One can see the “rule,id” that triggered the alert, the rule level (5) and more importantly the MITRE ATT&CK ID (T1110.001) and (T1021.004) with the tactic listed as “Credential access, lateral movement”. While the technique used is “password guessing”. If we add to this alert the first element which is the time, we can say that A SIEM tool provides us with key elements to confirm an incident as false positive or true positive, directing the analyst’s next action. Making it essential to SOC Threat detection activities.

2- WAZUH SIEM EVENTS DASHBOARD

After deployment, you can select an endpoint to check its events in the SIEM dashboard

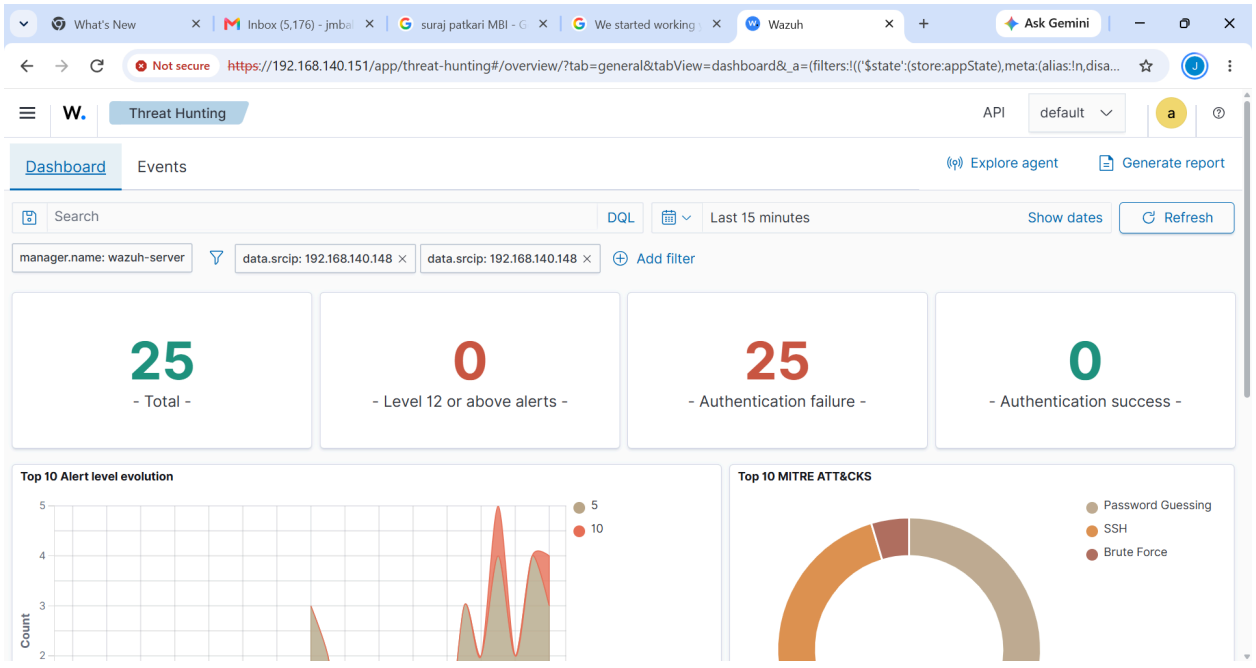


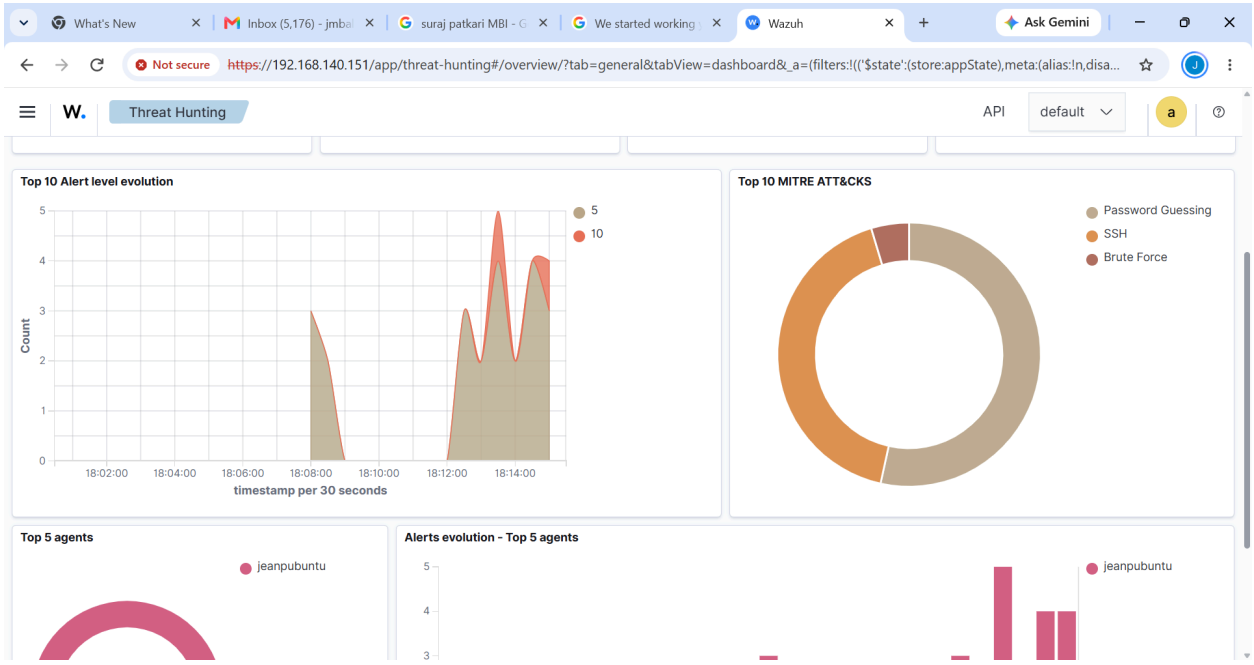


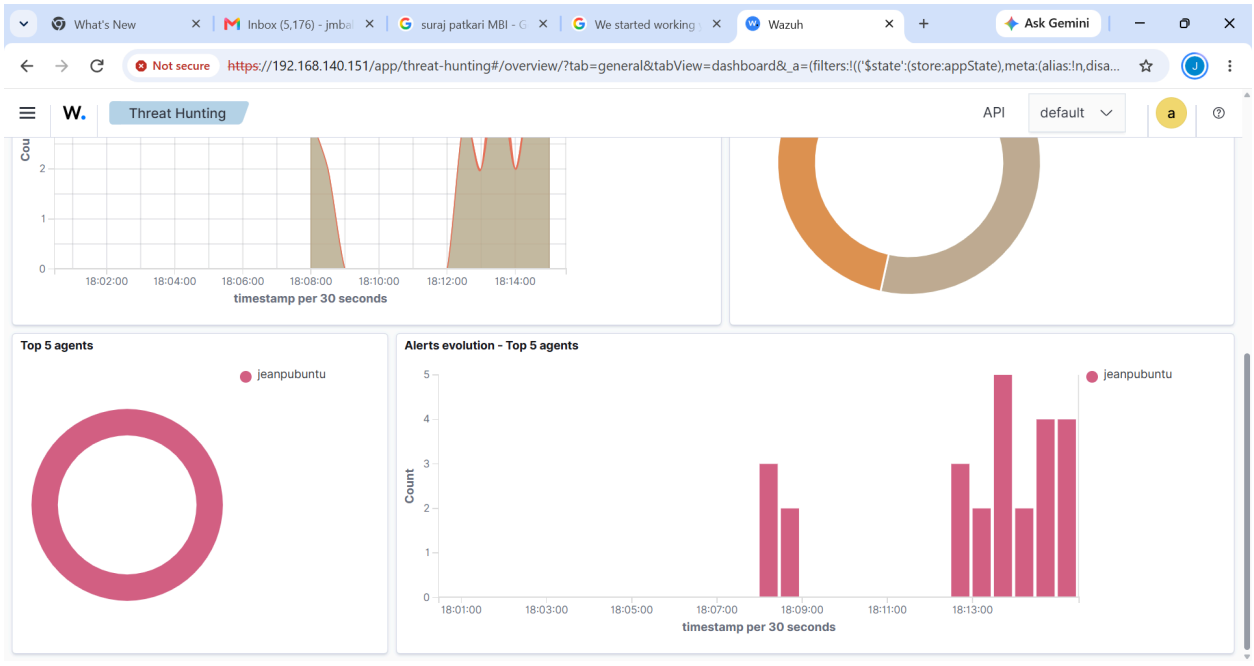
c- Screenshots of events list with their timeline.

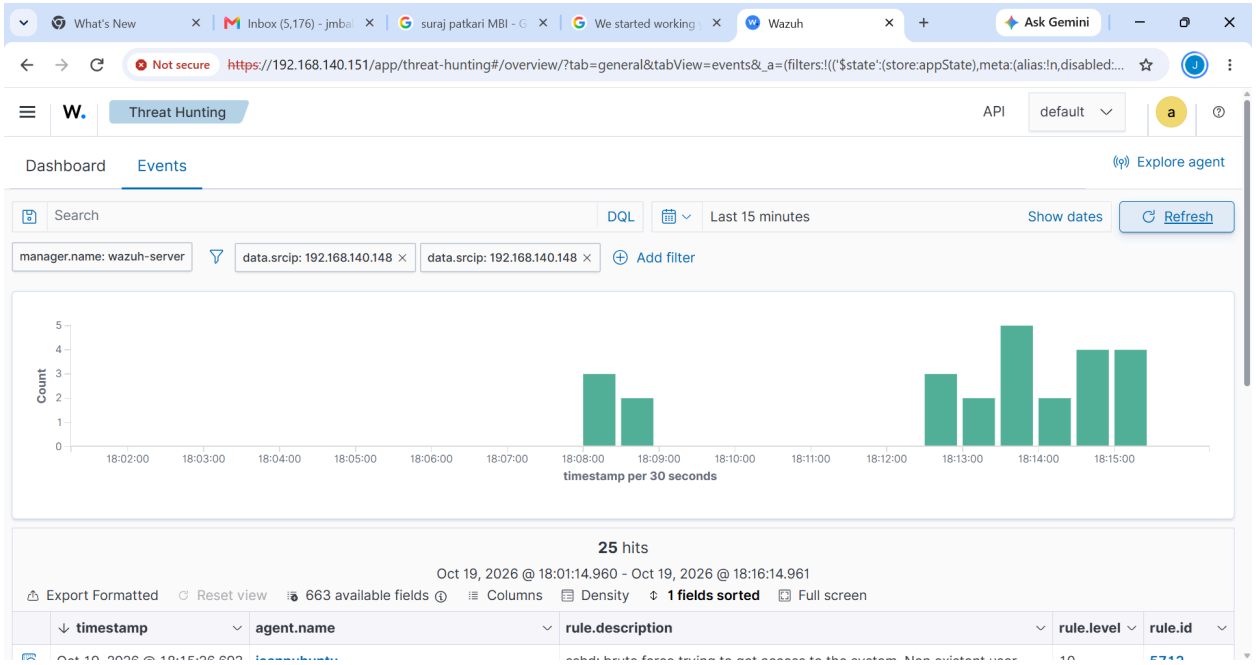
In the screenshots above we can see the alerts list with categories like timestamp, agent name, rule description, rule level and rule ID. Each of these values gives us key information about the alert, that will help us prioritize our triage and action.

Since our lab is about a SSH connection attack, the authentication rule has been triggered and the subsequent alerts populated in the SIEM dashboard as shown in the screenshots below



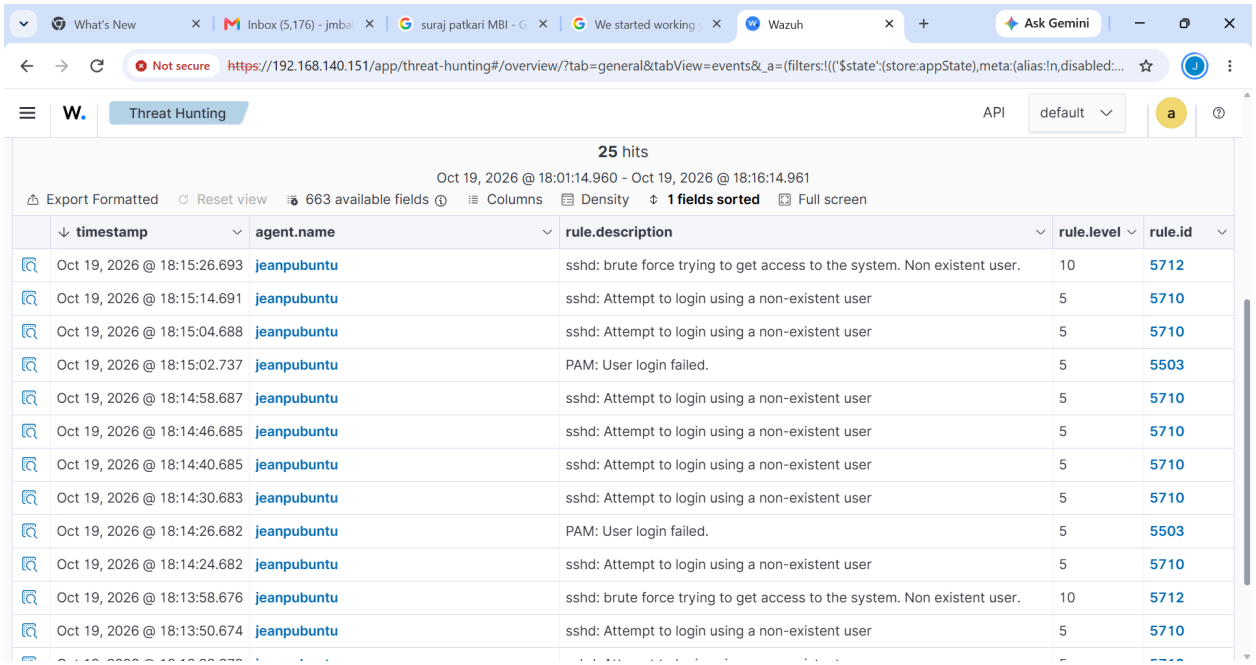

What's New
Inbox (5,176) - jmba
suraj patkari MBI - G
We started working
Wazuh
Ask Gemini
Not secure
https://192.168.140.151/app/threat-hunting#/overview/?tab=general&tabView=dashboard&_a=(filters:!(('$state':(store:appState),meta:(alias:!n,disa...
Threat Hunting
API
default
a
Top 10 Alert level evolution
Count
timestamp per 30 seconds
5
10
Top 10 MITRE ATT&CKS
Password Guessing
SSH
Brute Force
Top 5 agents
jeanpubuntu
Alerts evolution - Top 5 agents
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https://192.168.140.151/app/threat-hunting#/overview/?tab=general&tabView=dashboard&_a=(filters:!(('$state':(store:appState),meta:(alias:!n,disa...
Threat Hunting
API
default
a
timestamp per 30 seconds
Top 5 agents
jeanpubuntu
Alerts evolution - Top 5 agents
Count
jeanpubuntu

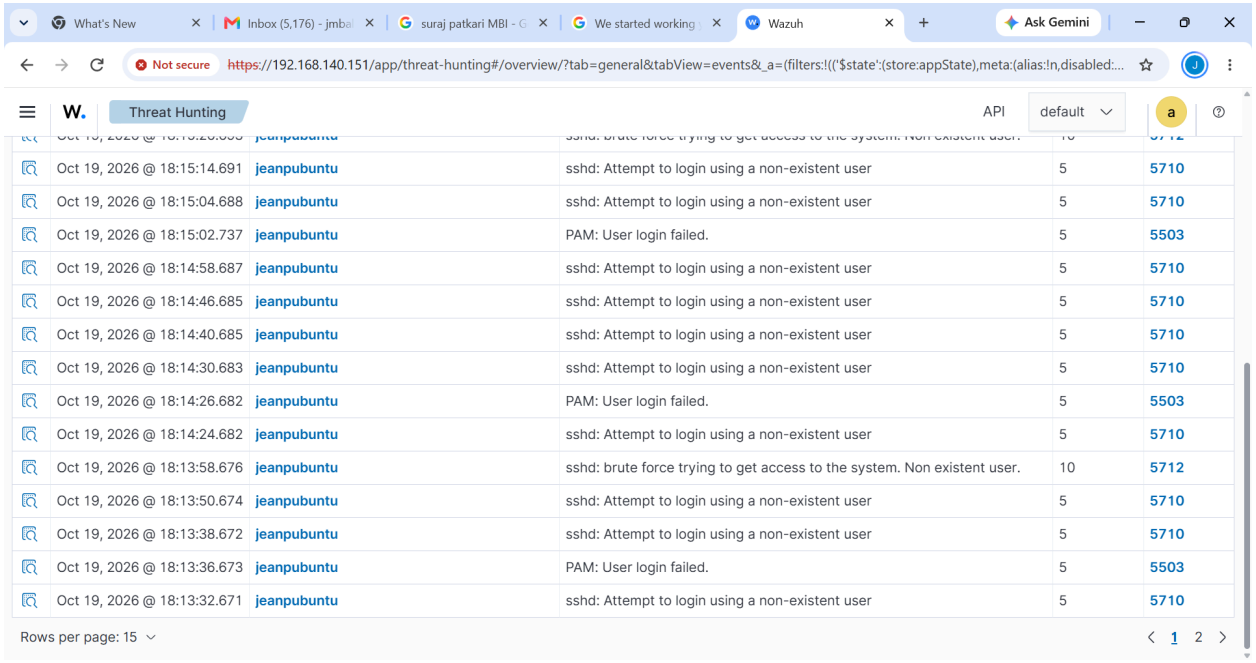


The above screenshots showed the dashboard information of the attack with the description graphs of Top 10 alerts, the number of attempts(25), the status as failed attempts(25), successful(0). Then we have the timestamp shown in seconds, indicating a potential brute force attack. That needed to be confirmed by initial analysis of patterns on the list and timestamp graph of occurrence, where we can see a spike in activity around the time it happened.

The screenshots below confirmed the rule triggered and the rule was elevated to rule (10) possible brute force attack logged by the SIEM tool:

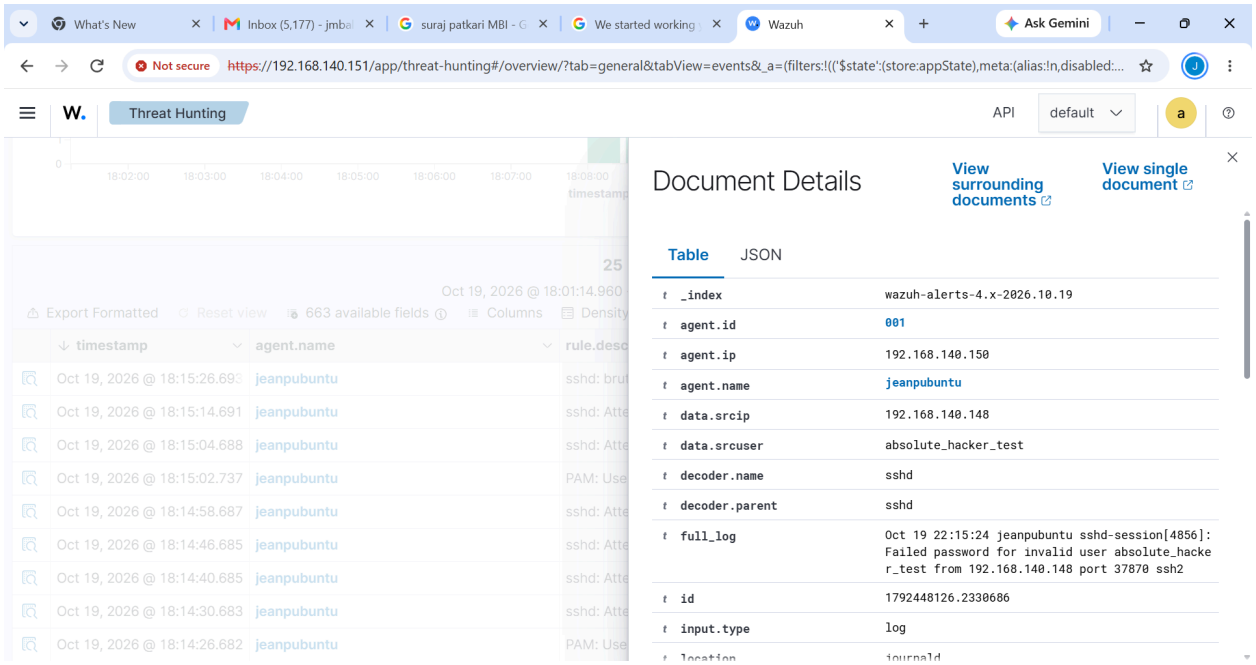


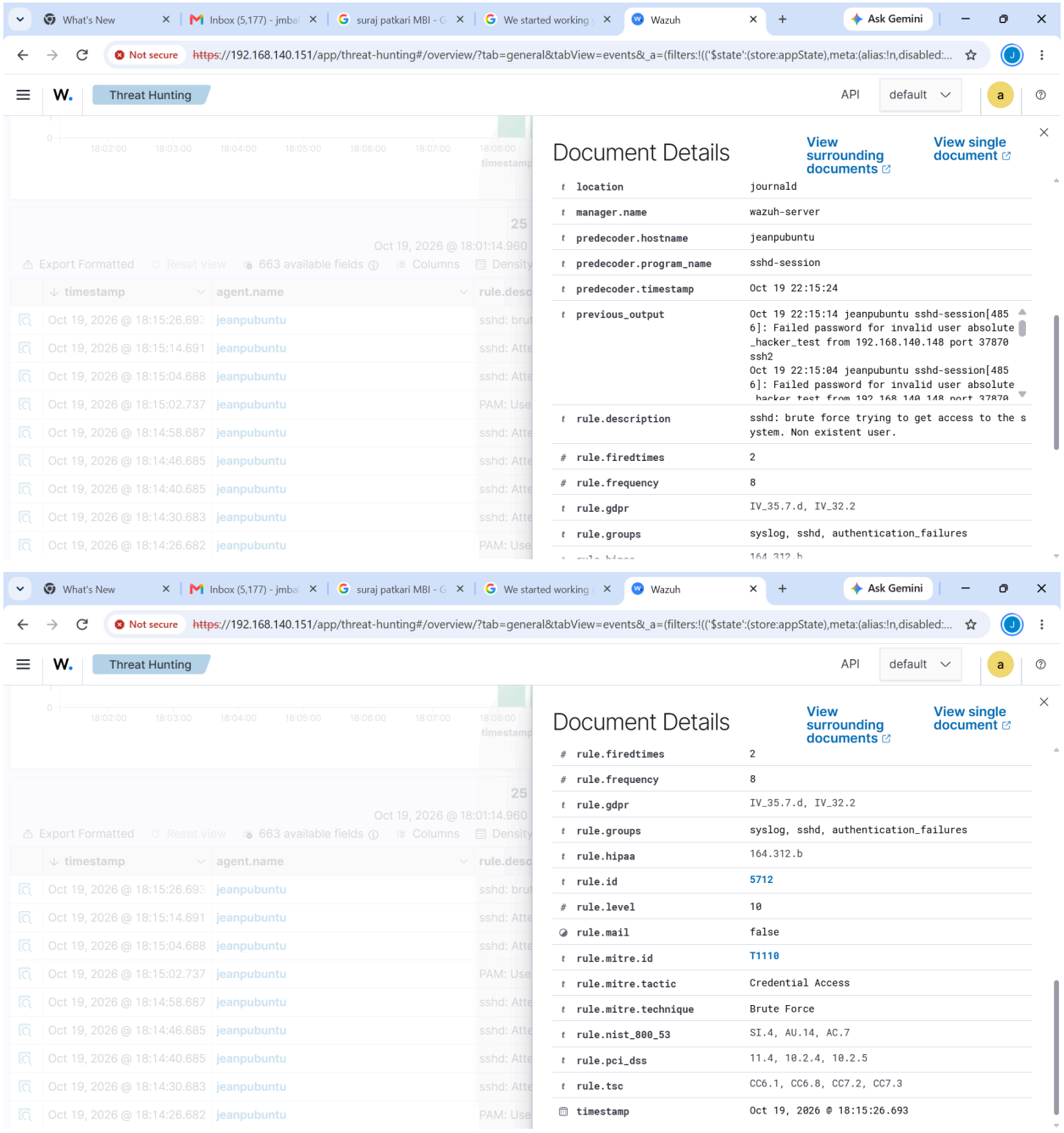
timestamp	agent.name	rule.description	rule.level	rule.id
Oct 19, 2026 @ 18:15:26.693	jeanpubuntu	sshd: brute force trying to get access to the system. Non existent user.	10	5712
Oct 19, 2026 @ 18:15:14.691	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:15:04.688	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:15:02.737	jeanpubuntu	PAM: User login failed.	5	5503
Oct 19, 2026 @ 18:14:58.687	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:14:46.685	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:14:40.685	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:14:30.683	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:14:26.682	jeanpubuntu	PAM: User login failed.	5	5503
Oct 19, 2026 @ 18:14:24.682	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:13:58.676	jeanpubuntu	sshd: brute force trying to get access to the system. Non existent user.	10	5712
Oct 19, 2026 @ 18:13:50.674	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710



Oct 19, 2026 @ 18:15:14.691	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:15:04.688	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:15:02.737	jeanpubuntu	PAM: User login failed.	5	5503
Oct 19, 2026 @ 18:14:58.687	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:14:46.685	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:14:40.685	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:14:30.683	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:14:26.682	jeanpubuntu	PAM: User login failed.	5	5503
Oct 19, 2026 @ 18:14:24.682	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:13:58.676	jeanpubuntu	sshd: brute force trying to get access to the system. Non existent user.	10	5712
Oct 19, 2026 @ 18:13:50.674	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:13:38.672	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710
Oct 19, 2026 @ 18:13:36.673	jeanpubuntu	PAM: User login failed.	5	5503
Oct 19, 2026 @ 18:13:32.671	jeanpubuntu	sshd: Attempt to login using a non-existent user	5	5710

Rows per page: 15 1 2





This is proof that our WAZUH SIEM is working as expected. The event was logged correctly to allow the analyst to proceed with confidence to escalation for further action.
The WAZUH SIEM tool captured the activities and presented us with a correlation of the event that triggered an alert confirmed as an incident with a level 10 rule assessment mapped to the MITRE ATT&CK ID (T1110) for credential access and described as a brute force attack . In an enterprise environment, this alert needs to be escalated right away to L2 at this point and containment action along with stakeholder communication starting simultaneously. We also have the NIST-800-53 rule stated, the PCI DSS rule involved, with the timestamp of when it happened that will help us build a timeline in our report for escalation.